

Section : Biology

Q.1 The degradation of humus by microbes to release inorganic nutrients is known as-

1. Humification
2. Mineralisation
3. Fragmentation
4. Catabolism

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Which one of the given assisted reproductive technologies (ART) involves transfer of fertilized egg into the fallopian tube?

1. IUI
2. ZIFT
3. GIFT
4. ICSI

Options 1. 1

2. 2
3. 3
4. 4

Q.3 The arrangement of the nuclei in a normal embryo sac from the chalazal to micropylar end in dicots is -

1. 3+3+2
2. 2+3+3
3. 3+2+3
4. 4+1+3

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Which one of the following plants is the source of cannabinoids commonly used in drug abuse?

1. *Papaver somniferum*
2. *Cannabis sativa*
3. *Erythroxylum coca*
4. *Atropa belladonna*

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Three cells grouped together at the micropylar end and constitute _____ in the embryo sac.

1. Central cell
2. Polar cells
3. Egg apparatus
4. Antipodals

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Recombinant DNA technology involves the following steps in a specific sequence.

- (A) Culture of the host cell
- (B) Extraction of desired product
- (C) Isolation of desired DNA fragments

(D) Transformation of the recombinant DNA into the host cell

Arrange these sequences correctly and choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (C), (D), (A), (B)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.7 When Tt x Tt are crossed, how many percent of the offsprings show a recessive phenotype?

1. 25%
2. 50%
3. 75%
4. 100%

Options 1. 1

2. 2

3. 3

4. 4

Q.8 Which one of the following represents a palindromic sequence?

1. 5'-GAATTC-3' 3'-CTTAAG-5'
2. 5'-CCAATG-3' 3'-GAATCC-5'
3. 5'-CATTAG-3' 3'-GATAAC-5'
4. 5'-GATACC-3' 3'-CCTAAG-5'

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Which one of the following statement is incorrect about ecological pyramids?

1. Pyramids of energy is always upright.
2. Saprophytes are given top place in ecological pyramids.
3. Pyramids of number can be inverted in the terrestrial ecosystem.
4. It doesn't take into account the same species belonging to two or more trophic levels.

Options 1. 1

2. 2

3. 3

4. 4

Q.10 *Bacillus thuringiensis* produces proteins (Bt toxin) that kill certain insects. Identify the correct statements.

- (A) Bt toxin protein exists as an inactive form (protoxin) in *Bacillus thuringiensis*.
- (B) Bt toxin protein exists as an active form toxin in the insect gut.
- (C) The alkaline pH of the gut of an insect solubilize the toxin and converts in an inactive form.
- (D) The toxin is coded by a gene *cryIAc* named cry.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (A), (B), (C) and (D)
3. (A), (B), and (D) only
4. (A) and (B) only

Options 1. 1

2. 2

3. 3

4. 4

Q.11 Match List-I with List-II

List-I**Birth control methods**

- (A) Diaphragms
- (B) Lactational amenorrhea
- (C) Vasectomy
- (D) IUDs

List-II**Features**

- (I) Removal of vas deferens
- (II) Increase phagocytosis of sperms
- (III) Absence of menstrual cycle and ovulation following parturition
- (IV) Cover the cervix and blocks the entry of sperm

Choose the correct answer from the options given below:

- 1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- 2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
- 3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
- 4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.12 Arrange the following structures involved in embryogenesis in the correct sequence.

- (A) Zygote
- (B) Blastocyst
- (C) Blastomeres
- (D) Morula

Choose the correct answer from the options given below:

- 1. (A), (C), (D), (B)
- 2. (A), (C), (B), (D)
- 3. (B), (A), (D), (C)
- 4. (C), (B), (D), (A)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.13 Which one of the following hormones is essential for maintenance of endometrium?

- 1. TSH
- 2. Estrogens
- 3. Progesterone
- 4. Relaxin

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.14 Which one of the following pyramids of biomass is generally inverted in the ecosystems?

- 1. Forest
- 2. Grassland
- 3. Sea
- 4. Desert

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.15 Which one of the following diseases is non-infectious?

- 1. Cholera
- 2. Pneumonia
- 3. Tuberculosis
- 4. Elephantiasis

Options 1. 1

2. 2

3. 3

4. 4

Q.16 Which one of the following disorders is a sex linked recessive disease?

1. Sickle-cell anaemia
2. Haemophilia
3. Phenylketonuria
4. Thalassemia

Options 1. 1

2. 2

3. 3

4. 4

Q.17 What is the primary function of the aeration tanks in the secondary treatment of sewage?

1. To remove solid waste through filtration
2. To allow the growth of aerobic microbes that consume organic matter
3. To reduce the BOD of the effluent using chemicals
4. To settle the activated sludge

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Sex determination in grasshopper is of -

1. XX-XY type
2. XX-XO type
3. ZZ-ZO type
4. ZZ-ZW type

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Which one of the following diagnostic techniques is based on the principle of antigen-antibody interaction?

1. Recombinant DNA technology
2. Polymerase Chain Reaction
3. Enzyme-linked immuno-sorbent assay
4. Gel electrophoresis

Options 1. 1

2. 2

3. 3

4. 4

Q.20 The biodiversity is _____ at lower altitudes and _____ with increasing higher latitudes.

1. Highest, Decreases
2. Lowest, Increases
3. Highest, Increases
4. Lowest, Decreases

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Select the correct sequence for transport of sperm in the male reproductive system.

- (A) Urethra
(B) Urethra meatus

(C) Ejaculatory duct

(D) Vas deferens

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (D), (C), (A), (B)
3. (D), (C), (B), (A)
4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Which of the following techniques are commonly used for early detection of cancer?

(A) Biopsy and histopathological studies

(B) Radiography (x-rays)

(C) MRI (Magnetic Resonance Imaging)

(D) Electrocardiogram (ECG)

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D) only
2. (A), (B) and (D) only
3. (A), (B) and (C) only
4. (A), (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.23 *Agrobacterium tumefaciens*, is able to deliver a piece of DNA known as _____ to transform normal plant cell into a tumor cell.

1. T-DNA
2. Z-DNA
3. B-DNA
4. C-DNA

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Which one of the followings is an indirect measure of the organic matter present in water?

1. pH level
2. Nitrogen content
3. Turbidity
4. BOD

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Identify the correct statements for adenosine deaminase (ADA) deficiency.

(A) It is caused by the failure of synthesis of an enzyme, adenosine deaminase.

(B) It leads to inherited immunodeficiency disorder, severe combined immunodeficiency (SCID).

(C) It is attributed to a mutant ADA gene.

(D) Its treatment can be done by introducing ADA cDNA into the hepatocytes.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Match List-I with List-II

List-I

Microorganisms

(A) *Lactobacillus*

(B) *Saccharomyces cerevisiae*

(C) *Propionibacterium sharmanii*

(D) *Clostridium butylicum*

List-II

Commercial Uses

(I) Use for the production of butyric acid

(II) Use for production of swiss cheese

(III) Use for making bread

(IV) Improves nutritional quality of milk by increasing vitamin B12

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
4. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.27 How many base pairs are present in one complete turn of DNA?

1. 10
2. 5
3. 7
4. 12

Options 1. 1

2. 2

3. 3

4. 4

Q.28 Match List-I with List-II

List-I

Terms

(A) Selectable marker

(B) Origin of replication

(C) Exonuclease

(D) Recognition site

List-II

Features

(I) Site that commonly used for cutting the DNA by restriction enzyme

(II) Removes nucleotides from the ends of the DNA

(III) Helps in identification and elimination of non-transformants

(IV) The sequence responsible for controlling the copy number of linked DNA

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.29 Which one of the followings cells are attacked by HIV?

1. Cytotoxic cells
2. B-cells
3. T-helper cells
4. T-memory cells

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.30 Human protein, α -1-antitrypsin is used for the treatment of-

- 1. Cystic fibrosis
- 2. Phenylketonuria
- 3. Emphysema
- 4. Alzheimer's disease

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.31 Which one of the following hormones is not secreted by the placenta?

- 1. hCG
- 2. hPL
- 3. Progestogens
- 4. Relaxin

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.32 During allergic reactions, which of the following are secreted?

- (A) Histamines
- (B) Interferons
- (C) Serotonin
- (D) Pyrogens

Choose the *correct* answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (B) and (C) only
- 3. (A), (B), (C) and (D)
- 4. (A) and (C) only

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.33 A specific recognition sequence identified by endonuclease to make cuts at specific positions within the DNA is -

- 1. Poly (A) sequence
- 2. Palindromic nucleotide sequence
- 3. Okazaki sequence
- 4. Degenerate primer sequence

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.34 Identify the correct combination of statements for sacred groves.

- (A) These are in-situ conservation strategies.
- (B) These are ex-situ conservation strategies.
- (C) These are found in the Khasi and Jaintia hills in Meghalaya.
- (D) These hold religious values.

Choose the correct answer from the options given below:

- 1. (A), (C) and (D) only
- 2. (A) and (D) only
- 3. (A) and (C) only
- 4. (B), (C) and (D) only

Options 1. 1

- 2. 2

3. 3

4. 4

Q.35 Which one of the followings is not a common mechanical barrier method for birth control?

1. Diaphragms
2. Vaults
3. Implants
4. Cervical caps

Options 1. 1

2. 2

3. 3

4. 4

Q.36 Arrange the following steps of curd formation from milk.

- (A) Coagulation and partial digestion of milk protein
 - (B) Growth of starter culture
 - (C) Growth of lactic acid bacteria (LAB) in milk
 - (D) A small amount of curd is added as an inoculum to fresh milk
- Choose the correct answer from the options given below:

1. (B), (D), (C), (A)
2. (D), (B), (C), (A)
3. (B), (D), (A), (C)
4. (C), (D), (B), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Match List-I with List-II

List-I

List-II

Pathogens

Diseases

- | | |
|-----------------------------------|--------------------|
| (A) <i>Wuchereria bancrofti</i> | (I) Ringworms |
| (B) Rhino Virus | (II) Elephantiasis |
| (C) <i>Haemophilus influenzae</i> | (III) Common Cold |
| (D) <i>Epidermophyton</i> | (IV) Pneumonia |

Choose the correct answer from the options given below:

1. (A) – (II), (B) – (III), (C) – (IV), (D) – (I)
2. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
3. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
4. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Arrange the following steps of genetically engineered insulin production.

- (A) Isolation and purification of the insulin gene from beta cells of the pancreatic islets
 - (B) Expression of gene in host cell
 - (C) Joining the two polypeptides by disulfide bonds
 - (D) Cloning of insulin gene in *Escherichia coli* cloning host cell
- Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (D), (B), (C)
3. (A), (B), (D), (C)
4. (A), (C), (D), (B)

Options 1. 1

2. 2

3. 3

4. 4

Q.39 Which one of the following plays an important role in guiding the entry of pollen tube into the synergid?

1. Filiform apparatus
2. Antipodals
3. Polar nuclei
4. Egg cell

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Match List-I with List-II

List-I

Scientists

(A) Frederick Griffith

(B) Francois Jacob and Jacques
Monod

(C) Har Gobind Khorana

(D) Meselson and Stahl

List-II

Discoveries

(I) Genetic Code

(II) Semi-conservative mode of DNA
replication

(III) Transformation

(IV) *Lac* operon

Choose the correct answer from the options given below:

1. (A) – (II), (B) – (I), (C) – (III), (D) – (IV)
2. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
3. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)
4. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the following passage carefully and answer the given questions.

Theory of Evolution

The evolution of life is a gradual process through which simple organisms are transformed into complex life forms over billions of years. It involves genetic changes, adaptation to the environment, and the emergence of diversity in species. Fossil records provide evidence of extinct life forms and transitional stages. Evolution is influenced by factors such as mutation, genetic recombination, and environmental pressure, shaping the biodiversity we see today. Charles Darwin and Alfred Russel Wallace were pioneering naturalists who developed the theory of evolution by natural selection.

The study of fossils in different sedimentary layers indicates _____.

1. Geographical periods in which they were existed
2. Geological periods in which they were migrated
3. Geological periods in which they were existed
4. Geographical periods in which they were migrated

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the following passage carefully and answer the given questions.

Theory of Evolution

The evolution of life is a gradual process through which simple organisms are transformed into complex life forms over billions of years. It involves genetic changes, adaptation to the environment, and the emergence of diversity in species. Fossil records provide evidence of extinct life forms and transitional stages. Evolution is influenced by factors such as mutation, genetic recombination, and environmental pressure, shaping the biodiversity we see today. Charles Darwin and Alfred Russel Wallace were pioneering naturalists who developed the theory of evolution by natural selection.

According to Darwin, the term 'Fitness' in the context of natural selection refers to:

1. Physical strength of an organism
2. The ability of an organism to adapt to any environment
3. The reproductive success of an individual or population
4. Survival without reproduction

Options 1. 1

2. 2
3. 3
4. 4

Q.43 Read the following passage carefully and answer the given questions.

Theory of Evolution

The evolution of life is a gradual process through which simple organisms are transformed into complex life forms over billions of years. It involves genetic changes, adaptation to the environment, and the emergence of diversity in species. Fossil records provide evidence of extinct life forms and transitional stages. Evolution is influenced by factors such as mutation, genetic recombination, and environmental pressure, shaping the biodiversity we see today. Charles Darwin and Alfred Russel Wallace were pioneering naturalists who developed the theory of evolution by natural selection.

The voyage of the H.M.S. Beagle was significant in the development of Darwin's theory of evolution as-

1. It allowed Darwin to study fossilized remains of ancient organisms in museums.
2. It provided direct observations of artificial selection in controlled environment.
3. It helped Darwin to identify patterns of adaptive radiation of species.
4. It gave Darwin access to laboratory experiments that confirmed genetic mutation.

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the following passage carefully and answer the given questions.

Theory of Evolution

The evolution of life is a gradual process through which simple organisms are transformed into complex life forms over billions of years. It involves genetic changes, adaptation to the environment, and the emergence of diversity in species. Fossil records provide evidence of extinct life forms and transitional stages. Evolution is influenced by factors such as mutation, genetic recombination, and environmental pressure, shaping the biodiversity we see today. Charles Darwin and Alfred Russel Wallace were pioneering naturalists who developed the theory of evolution by natural selection.

Alfred Russel Wallace, a naturalist, arrived at conclusions similar to Darwin while working in:

1. Galapagos island
2. Malay archipelago
3. Amazon rainforest
4. Australian outback

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the following passage carefully and answer the given questions.

Theory of Evolution

The evolution of life is a gradual process through which simple organisms are transformed into complex life forms over billions of years. It involves genetic changes, adaptation to the environment, and the emergence of diversity in species. Fossil records provide evidence of extinct life forms and transitional stages. Evolution is influenced by factors such as mutation, genetic recombination, and environmental pressure, shaping the biodiversity we see today. Charles Darwin and Alfred Russel Wallace were pioneering naturalists who developed the theory of evolution by natural selection.

Which one of the following statements is not a connotation of the theory of special creation?

1. All living organisms that we see today were created as such.
2. The diversity of life has remained constant since creation.
3. Earth is approximately 4000 years old.
4. Life forms evolved gradually over millions of years.

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the following passage carefully and answer the given questions.

Productivity in the Ecosystem

The flow of energy through a terrestrial ecosystem starts with the harnessing of sunlight by autotrophs. The radiant energy is converted by photosynthesis to organic compounds is known as productivity. The productivity is expressed in units of energy per unit of area per unit time. The amount of organic matter present at any given time is biomass. The productivity of any ecosystem is affected by many parameters like weather conditions, nutrient availability etc. The productivity of any ecosystem does vary seasonally and yearly.

The unit for expressing productivity is -

1. $(\text{Kcal m}^{-2}) \text{ yr}^{-1}$
2. $(\text{Kcal m}^{-2}) \text{ yr}^{-2}$
3. $(\text{Kcal m}^{-1}) \text{ yr}^{-1}$
4. $(\text{Kcal m}^2) \text{ yr}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.47 Read the following passage carefully and answer the given questions.

Productivity in the Ecosystem

The flow of energy through a terrestrial ecosystem starts with the harnessing of sunlight by autotrophs. The radiant energy is converted by photosynthesis to organic compounds is known as productivity. The productivity is expressed in units of energy per unit of area per unit time. The amount of organic matter present at any given time is biomass. The productivity of any ecosystem is affected by many parameters like weather conditions, nutrient availability etc. The productivity of any ecosystem does vary seasonally and yearly.

Identify the correct statement in relation to productivity in an aquatic ecosystem.

1. The GPP is always less than NPP
2. The GPP is always more than NPP
3. The GPP is always equal to NPP
4. No relation between GPP and NPP

Options 1. 1

2. 2
3. 3
4. 4

Q.48 Read the following passage carefully and answer the given questions.

Productivity in the Ecosystem

The flow of energy through a terrestrial ecosystem starts with the harnessing of sunlight by autotrophs. The radiant energy is converted by photosynthesis to organic compounds is known as productivity. The productivity is expressed in units of energy per unit of area per unit time. The amount of organic matter present at any given time is biomass. The productivity of any ecosystem is affected by many parameters like weather conditions, nutrient availability etc. The productivity of any ecosystem does vary seasonally and yearly.

The biomass available for consumption by herbivores and decomposers is known as-

1. Gross Primary Productivity
2. Net Primary Productivity
3. Secondary Productivity
4. Standing Biomass

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the following passage carefully and answer the given questions.

Productivity in the Ecosystem

The flow of energy through a terrestrial ecosystem starts with the harnessing of sunlight by autotrophs. The radiant energy is converted by photosynthesis to organic compounds is known as productivity. The productivity is expressed in units of energy per unit of area per unit time. The amount of organic matter present at any given time is biomass. The productivity of any ecosystem is affected by many parameters like weather conditions, nutrient availability etc. The productivity of any ecosystem does vary seasonally and yearly.

The amount of organic matter accumulated in an ecosystem at a given time is known as-

1. Standing state
2. Standing crop
3. Standing biomass
4. Standing productivity

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the following passage carefully and answer the given questions.

Productivity in the Ecosystem

The flow of energy through a terrestrial ecosystem starts with the harnessing of sunlight by autotrophs. The radiant energy is converted by photosynthesis to organic compounds is known as productivity. The productivity is expressed in units of energy per unit of area per unit time. The amount of organic matter present at any given time is biomass. The productivity of any ecosystem is affected by many parameters like weather conditions, nutrient availability etc. The productivity of any ecosystem does vary seasonally and yearly.

The Net Primary Productivity of an ecosystem are represented by:

1. $NPP = GPP + \text{Respiration}$
2. $NPP = GPP - \text{Respiration}$
3. $NPP = \text{Radiant Energy} - GPP$
4. $NPP = \text{Radiant Energy} + GPP$

Options 1. 1

2. 2
3. 3
4. 4